

Topic: Tackling missing data, outliers, or imbalanced datasets in ML

Title: Evaluating Support Vector Machines (SVMs) as base learners within ensemble models for performance working with imbalanced datasets

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Project Proposal:

The challenge of class imbalance is well-documented in clinical informatics, where "out-of-the-box" classifiers often fail to capture high-stakes minority samples. Liu et al. (2022) demonstrate that ensemble algorithms are essential for screening rare conditions like Aortic Dissection, as they prevent the majority class from overshadowing the clinical signal. Recent advancements by Albahli & Yar (2025) and Ganie et al. (2025) emphasize that integrating explainable AI within ensemble frameworks, such as ST-DenseNet, significantly improves the adoptability of cardiovascular diagnostic predictions. Specific to Support Vector Machines, Wang et al. (2017) established that the synergy between data-level resampling (e.g., SMOTE) and algorithmic bagging yields superior robustness compared to standalone SVMs. Furthermore, research by Liu et al. (2020) and Rabie & Saleh (2024) suggests that complex configurations, including Cost-Sensitive Learning and hierarchical stacking, are necessary to transition from theoretical models to professional-grade diagnostic decision support systems. Finally, explainability is also coming to the forefront of machine learning-informed diagnostics, bringing in tools such as SHAP (SHapley Additive exPlanations) to inform users on what factors led the algorithm to its decision (2026).

The core objective is to conduct a comparative analysis between components of ensemble models to develop an understanding of how the performance of an ensemble method is affected by changes to the components of the ensemble. We will implement data-level

interventions, including undersampling and Synthetic Minority Over-sampling Technique (SMOTE). SVM will be used as the base learner, and compared to logistic regression as a base learner. The performance of the ensemble will be further observed when the weight of the minority class is manipulated. Both SVMs and logistic regression algorithms will also be evaluated outside of the ensemble. Evaluation will be conducted on representative medical data, prioritizing metrics such as Sensitivity (Recall), Area Under the Precision-Recall Curve (AUPRC), and the Matthews Correlation Coefficient (MCC) over standard accuracy. The specific datasets we have chosen to test on are Breast Cancer Wisconsin (Wolberg, 1990) and SPECT Heart (Cios, 2001) from the UCI Machine Learning Repository due to their imbalance nature and relevance to prior works. By transitioning from "out-of-the-box" single classifiers to engineered ensemble systems, this project replicates professional data science workflows where capturing high-stakes, minority-class samples is a critical requirement for diagnostic decision support.

Regarding specific code that needs to be written, the core task at hand is to study the performance of SVMs as a base learner within an ensemble method, and compare it to SVM performance outside of the ensemble, as well as to the performance of logistic regression as a base learner. Various class weightings will be explored in this effort as well. Secondly we will evaluate preprocessing techniques, which will have their own associated functions that will easily be able to be called within different "full stack" scripts.

Fall Back Plan:

In the event that the integration of multi-stage stacking or complex cost-sensitive ensemble architectures proves computationally prohibitive or fails to converge on the clinical datasets, the team will pivot to a controlled comparative study of modified single-learner SVMs versus standard Bagging ensembles.

Instead of the ideal goal of engineering a hybrid "Stacking" architecture, we will focus on a rigorous evaluation of "Class-Weighted SVMs" (algorithmic level) against "Balanced Bagging" (data level). We will continue to evaluate using the previously mentioned datasets from the UCI Machine Learning Repository. This ensures that even if the advanced ensemble engineering is unattainable, we still provide a definitive analysis of which intervention, algorithmic weighting or ensemble-based resampling, is most effective for SVMs in a clinical context.

Division of Labor

Dakota Wang

- **Data Engineering & Preprocessing:** End-to-end data pipeline includes cleaning the clinical datasets, handling missing value imputation, and implementing data-level interventions for class imbalance. Specifically, develop the pipeline for Synthetic Minority Over-sampling Technique (SMOTE) and random undersampling to create balanced training sets.
- **Statistical Evaluation:** Lead the performance analysis. Will implement the scoring suite to calculate non-accuracy-based metrics, including Area Under the Precision-Recall Curve (AUPRC), Matthews Correlation Coefficient (MCC), and Recall (Sensitivity), ensuring the models are validated for clinical utility.

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- **Ensemble Architecture Design:** Responsible for the high-level architectural framework. In addition to designing and implementing the Stacking wrappers that house the SVM base learners.
- **Base Learner Optimization:** Focus on the implementation and fine-tuning of the Support Vector Machine (SVM) models. This involves optimizing the maximum-margin

hyperplane and implementing Cost-Sensitive Learning by manually adjusting the class-weight parameters to penalize minority-class misclassifications.

References:

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